SAMPLE

Building the Mixed Reality of Tomorrow

Building the Blended Worlds of Tomorrow

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Introduction

Mixed reality is the quietest of the decade's large computing ambitions, and the least understood of its headline terms -- described at various times as a headset category, a marketing compromise between two older words, a consumer product that came and went, and a laboratory dream, when it is in fact none of those things and, in its working definition, a precise engineering contract that the decade's hardware is only now beginning to be able to honor. This volume treats mixed reality the way the series treats every technology it covers: as an engineered system with a definition that can be checked, an anatomy that can be assembled, failure modes that can be named in advance, and an operating discipline a builder can run -- because the blend that the demonstrations describe is a construction project, and construction projects do not fail for lack of spectacle. They fail for lack of structure.

What mixed reality actually is, in the definition this book will defend: a computed environment in which the virtual and the physical coexist in the same coordinate space, observe the same room, and act on each other -- the machine holding a continuously updated model of the real environment so that virtual things can be placed in it, occluded by it, lit by it, collided with, and changed by it, while the person inside acts on both layers with one body and one grammar. The contract's load-bearing clause is the mutual one: the room is modeled, the model is live, and the virtual behaves as if it belongs -- which is what separates mixed reality from its two neighbors in the series' earlier volumes. Augmented reality, treated in Volume 27, pins information to the world; its subject is the anchored layer, the world seen through notes and markers, and its hardest problem is registration. The metaverse, treated in Volume 28, builds persistent shared worlds; its subject is the place, and its hardest problem is world-hood. Mixed reality's subject is the merge itself: not what the world is annotated with, nor what world is built, but the seam where a computed thing and a physical room are made to obey one another -- and the seam, in engineering as in building, is where the structure either holds or fails.

The volume follows the series pattern. It opens with the definition and the contract -- what mixed reality precisely is, the three clauses the word commits to, the impostors that attach themselves to it, and why the exclusions matter more than the inclusions. It assembles the anatomy: the room's live model, the geometry and semantics every believable blend must carry; the capture-to-display eye, the passthrough pipeline whose latency and honesty set the comfort envelope; the laws the virtual must obey -- occlusion, collision, light, sound -- and the authored exceptions that make the medium expressive; the body's grammar of hands, gaze, and voice; the anchors that give the blend a memory across sessions; the shared room, where co-presence makes the blend social; the authoring discipline that makes hybrid scenes buildable and testable; the deployment envelope, where the blend meets working life and its economics; and -- as the series does in every volume -- the operating program, the five lines of records, calendar, budget, training, and handover that convert a demonstration into an institution.

The promises are the series' promises. No vendor names, no expiring statistics, no forecasts dressed as facts -- because the structure is the subject, and structure is the only layer that survives the product cycle. Mixed reality has already burned through waves of expectation and dismissal, with hardware generations arriving and being retired while the disciplines underneath compounded quietly; the material in this book is drawn from what remains when each wave clears, and it will still be true when the next one arrives. Read it as the builder's structural guide to the hardest interface problem in computing: making a computed thing and a physical room answerable to one another, in real time, with a human body standing inside the result -- because the body, as the series has argued in every volume it has touched, is the harshest auditor in engineering, and it does not read press releases. That audit is the subject here, and the book begins where every honest contract begins: with the definition, and what the word commits to.

Chapter 1: The Definition and Its Contract

Any technology that arrives in a borrowed word must eventually settle the word's debts, and few terms in modern computing have been stretched further than this one. Mixed reality -- a coinage born in a laboratory, aged through decades of demos, and finally adopted as a marketing halfway-house between two older terms -- carries a specific, checkable meaning, and the chapter's first job is to make the specification exact: mixed reality is a computed environment where virtual and physical content occupy one coordinate space and act on each other, mediated by a machine holding a continuously refreshed model of the real environment.

The virtual is placed in the room, not merely over it. The room occludes the virtual, reflects light onto it, and blocks its paths. The virtual returns the favor: it can mark the room, measure it, mask it, and reorganize what the person sees of it. And the person inside acts on both layers with one continuous body -- one grammar of hands, gaze, and

voice -- never switching modes between a real act and a computed one. The contract stops there, and this chapter stress-tests every clause, because what looks like industry confusion about mixed reality is rarely confusion about the technology. It is disagreement about the contract -- a seller demonstrating one clause, a buyer hearing another, both gesturing at the same goggles while describing different products.

A definition is not a menu. It has three clauses, and each is present because removing it yields a known, lesser product already on the market -- and those lesser products are exactly what keeps getting sold under the name. The first clause is co-location: virtual and physical share a single coordinate space.
Drop it and you have a picture-in-picture -- computed content overlaid on a camera feed, aligned to nothing, obeying nothing, which is a screen effect, valuable and mature and not a blend. The second clause is mutual action: the two layers affect each other, through occlusion, collision, lighting, and the person's shared manipulation. Drop it and you have augmented reality's simpler half -- annotation pinned to the world, the anchored layer the series treated in Volume 27 -- where the room is a bulletin board and never a participant. The third clause is the live model: the machine's understanding of the environment is continuously updated while the person is inside it, because rooms are not static -- doors open, people walk through, furniture moves, light changes -- and a blend computed against a stale model of the room is a blend against a room that no longer exists. Drop it and you have a pre-scanned diorama, a frozen museum of a moment. The three clauses together define the merge, and the merge is the engineering problem: co-location demands world-locked registration, mutual action demands a physics both layers respect, and the live model demands perception that runs faster than the room changes.

The famous spectrum that places the real environment at one pole and the fully virtual at the other, with intermediate blends strung between, is the field's most useful diagram and its most abused one. Useful, because it establishes the honest frame: mixed reality is not a product category but a range of positions on a continuum, and any given deployment occupies a point on that range -- how much of the world is the person seeing, how much is computed, and where exactly the seam sits. Abused, because the diagram tempts two errors. The first is the ladder error: the belief that positions on the continuum are rungs of maturity, that a deployment closer to the virtual pole is more advanced than one closer to the real pole. The truth is the opposite of that -- each position is a design decision, not a grade. A technician guided through a physical machine needs the real pole, and the blend should be nearly all room; a pilot training on a procedure needs the virtual pole, and the blend should be nearly all computation. The second is the seam error: the belief that the continuum is smooth, that moving along it is a matter of degree. It is not. The seam -- the point where the physical hands off to the virtual -- is where the engineering lives, and the continuum's positions are separated by the seam's placement, which is a structural choice with structural consequences. The builder's discipline is to choose the position deliberately, place the seam consciously, and never let a marketing diagram make the decision.

The buyer's first line of defense is the catalog itself: the label has been pinned on at least four different products, and every pinning moved money. The camera-equipped headset: video relayed through a display, however good the display, is a window, and a window is not a merge -- unless the world being passed through is modeled, can be occluded, and can be acted on, it is the first clause's picture-in-picture, sold in goggles.
The overlay demo: a virtual object floating convincingly in a well-lit, pre-scanned, carefully staged room, which is the frozen diorama clause, demonstrating the blend against the one room it was built for and implying all rooms. The session toy: a blend that evaporates when the goggles sleep, which abandons the memory question the medium exists to ask -- placed things that persist are the difference between an environment and an installation. Then the rebrand: platforms that took up the word while continuing to ship the session-scoped, single-room, single-user product they had always shipped, the definition repurposed as a trademark -- the opposite of what a definition is for. Every impostor is a genuine product with genuine uses; not one of them is the contract; and a procurement that cannot hold the catalog straight will buy the impostor, hit the seam at the worst possible moment, and pronounce the failure a verdict on the entire field.

A definition this strict pays for itself in three transactions, starting with evaluation. Measured against the three clauses, any product can be scored inside an afternoon: do virtual things share the room's coordinates, or only its screen?
Does the room act on them -- occlude, light, collide -- and do they act back on the room? Is the room's model live, or was the demonstration built against yesterday's room? The second transaction is engineering: the clauses are the work packages. Co-location is a registration and tracking problem, mutual action lives in simulation and rendering, the live model is a perception problem, and a program unable to say which team owns which clause has not yet begun. The third transaction is foretelling: the contract shows the builder which failure arrives first. For the naive it is the live model, because perception is the least glamorous discipline and the first item descoped.

Co-location fails second, when the tracking drifts and the placed thing slides off its anchor. Mutual action fails last and loudest, because it is the clause the demonstration promised the buyer -- the moment the virtual hand passes through the real table, the room itself testifies against the product. A builder who knows the clause order funds the disciplines in the order the contract will audit them.

Exclusion is the definition's gift, and four exclusions do most of the work. Mixed reality is not a display technology: displays are how the blend reaches the eyes, and they are the most visible layer and the least decisive one -- the contract can be met by a handheld device that models the room and acts in it, and violated by the finest headset ever manufactured. It is not virtual reality with cameras: virtual reality is a legitimate medium whose subject is the replacement of the world, and bolting a passthrough feed onto it produces, at best, a window; the merge is a property of the system's understanding, not of its optics. It is not the metaverse: the persistent shared world of Volume 28 is a place; mixed reality is a relationship -- a way physical rooms and computed things can have with each other, and a metaverse can be entered without any room, and a room can host a blend without any world. And it is not a consumer category: the contract is indifferent to who is inside it. The technician, the surgeon, the student, and the game player are all standing in the same engineering, and the discipline that serves one serves the others. Each exclusion protects a real boundary: between the medium and its hardware, between neighbors that share vocabulary, and between the builder's problem and the market's noise.

The structural fact that distinguishes mixed reality from every interface before it is this: for the first time, the physical environment is a first-class citizen of the computation. In every prior interface, the world was outside the system -- the screen showed the computation, the world held the screen, and the two never met except through the person's neck. In a blend, the room is inside the system: modeled as geometry, parsed for semantics, lit and re-lit by the renderer, consulted by the physics, and changed by the virtual in return. This inversion carries consequences the rest of the book develops. The room becomes state -- something the system must remember, update, and forget responsibly, with all the storage and privacy questions state implies. The room becomes a dependency -- the system's quality is bounded by the quality of its model of a place it did not build and cannot fully control. And the room becomes a collaborator -- the most powerful rendering, collision, and simulation engine in the building, running for free, whose behavior the computed layer must learn to respect rather than replace. The builder who treats the room as scenery builds a fragile thing; the builder who treats it as the system's largest component builds the medium the definition describes.

The contract's deepest clause is unwritten but enforced by the body: the blend must be believable, and believability is a budget, not a resolution. The person inside a merge maintains, mostly below awareness, a running account of whether the computed things belong in the room -- and the account is kept in the body's ledger, settled in the currency of comfort, attention, and trust. Every defect spends the budget: an object that slides a centimeter off its anchor spends attention; a shadow that falls the wrong way spends belief; a table the virtual hand passes through spends the whole account at once, because the body has no sense of humor about collision. The budget metaphor does real engineering work, so it is worth stating precisely. Believability is not accumulated by fidelity alone -- a high-polygon object that ignores the room is less believable than a simple object that obeys it, which is why the disciplines of this book are disciplines of obedience rather than of detail. Believability is spent unevenly -- the body forgives more in the periphery than at the center of gaze, more for a moment than for a minute, more in motion the person initiates than in motion imposed on them. And believability, once overspent, is not refunded by later correctness: the person who has seen a hand pass through a table will watch the next virtual object the way they watch a stage magician -- impressed, perhaps, but not present. The chapters that follow are, in effect, a budget manual: where the costs are, which defects spend the most, and how a builder stays solvent.

Boundary cases are where definitions are made or broken, and the honest treatment arbitrates three. The heads-up display that draws a line on the physical floor: annotation, pinned to the world -- augmented reality's anchored layer, until the line responds to the room's obstacles, at which point the room has begun to act on the computed and the boundary is crossed. The fully virtual environment scanned from a real one: a model of a room, however accurate, is a world -- the metaverse's territory -- until live sensors keep it synchronized with the room's changes, at which point the room is participating and the blend has begun. The remote collaborator projected into the physical office as a volumetric presence: a computed thing placed in a room -- the blend's territory, and among its most demanding cases, because the projected person is the one virtual object whose believability the audience audits socially as well as optically. The arbitrations share a principle: the boundary is not drawn at the device, nor at the display, nor at the degree of virtuality, but at the relationship. Wherever computed content and physical environment begin to hold one coordinate space and act on each other through a live model, the contract's subject is present -- and the book's disciplines apply, whatever the hardware wearing them.

For the reader holding a purchase order rather than a build plan, the contract compresses into a battery the seller can be asked to survive. Ask for the blend in an unprepared room: a product whose demonstration requires a pre-scanned, well-lit, obstacle-cleared stage is selling the frozen diorama. Ask what the virtual thing does when the room intervenes: a person walks between the display and the placed object, a door opens and the light changes, a table is dragged across the anchor -- the answer describes the mutual-action clause better than any specification. Ask what survives the sleep: whether placed content, room calibration, and learned semantics persist across sessions, which is the memory question the anchors chapter prices. Ask who owns the room's model: where the scan lives, who can read it, and what happens to it when the device is returned -- the privacy question the reconstruction chapter raises to first-class status. And ask what the body is asked to do: the interaction grammar, the comfort envelope, and the failure behavior when tracking is lost. A product that survives the battery honors the contract; a product that survives the demonstration but not the battery is an impostor, however sincere, and the difference will be discovered at the seam.

The definition's last service is what it declines to promise, and the mixed reality contract declines three promises marketing keeps making.
It refuses the replacement promise: the blend does not replace the world, and the product that hides the room is a virtual reality system wearing the wrong name -- the room is the medium's partner, and partners are not replaced. It refuses the universality promise: the blend is not for every task, and the discipline of Chapter 9 is largely the discipline of knowing which tasks want the room and which want the world -- the continuum's positions are choices, and a medium that claims every position has chosen none. And it refuses the inevitability promise: nothing about the contract is automatic, because every clause -- the shared space, the mutual action, the live model -- is funded engineering, and the funding must be renewed every generation of hardware. The exclusions are the gift because they convert the field from a hype cycle into a discipline: the builder who accepts them knows what to build, the buyer who accepts them knows what to audit, and the practitioner who accepts them knows what to practice. The chapters that follow are the audit's expansion -- the anatomy of the merge, the laws it obeys, the memory it needs, the sharing it affords, and the program that keeps it running.

The definition must also be defended against its friends -- the enthusiasts whose praise damages the field more than the skeptics' contempt, because praise lowers the guard. The friend who calls the merge indistinguishable from reality invites the audit the medium cannot yet pass and sets expectations the body will spend its budget failing; the honest claim is believability within an envelope, and the envelope is a design artifact. The friend who calls the medium inevitable invites the descope: if the blend is destiny, then the unglamorous clauses -- the live model, the persistence, the evaluation -- can be deferred, and they are always the ones the incidents come from. The friend who calls every room-scale computation mixed reality dissolves the definition into a mood, and the word that means everything protects nothing. Against all three, the defense is the same: the contract is strict because the medium is real -- strictness is how real engineering is honored, and the fields that refuse a strict definition are the fields that mistake their demos for their products. Mixed reality's disciplines are learnable, runnable, and durable; the best thing anyone can do for them is hold them to a standard.

Every chapter of this book expands a contract short enough for one sentence, worth writing on the whiteboard before the expansion starts: mixed reality is a computed environment where virtual and physical things maintain one shared, live model of one room and act on each other through it, one body inside.
Each phrase earns its place. A computed environment -- the system is software plus sensors, not a display plus content. Virtual and physical things -- both layers are participants, neither is a backdrop. One live model -- the room is understood continuously, or the merge is against a memory. One room -- the medium is located, which is what makes it different from the web and the metaverse alike. And one body inside -- the person is the shared citizen of both layers, the auditor of the merge, and the reason the whole construction exists. The sentence is the contract; the rest of the book is the clauses.

The contract's clauses meet at a specific place, and the builder who can name the seam can defend the definition against every ambiguity the market throws at it. The seam is the boundary where the physical and the computed exchange authority: on one side, the room's own truth -- light, matter, distance, consequence -- asserted by the world and unimprovable by engineering; on the other, the computed truth -- geometry, behavior, appearance -- asserted by the system and wholly owned by it. The seam's placement is the design act that Chapter 1's continuum described, and the act has three honest placements, each producing a different medium. The seam at the eye: the whole world is processed, and the computed layer's authority begins only at the pixels -- the head-mounted pipeline's placement, with the strictest budgets, because the entire room crosses the boundary every frame. The seam at the hand: the eye's view of the world is native, and the computed layer's authority begins

at the instrument -- the handheld window's placement, where the blend lives in a held rectangle and the room outside it is gloriously, safely real. The seam at the surface: the room is native everywhere, and the computed layer's authority begins where the light lands -- the projected window's placement, where the blend is public, fixed, and shared by everyone in the room with no device worn at all. Naming the seam converts arguments about hardware into arguments about authority, which is what they always were: the enthusiast asking why the field's devices are so heavy is asking a question about where the seam sits, and the buyer who answers it deliberately has made the single most consequential decision in the deployment before any budget is spent.

Definitions end in vocabularies, because a shared vocabulary is what a definition is for -- and the reader deserves, before the anatomy begins, the small dictionary the rest of the book will lean on, stated once and used consistently. The room: the physical environment inside the blend, first-class citizen and physics engine, whose authority the computed layer borrows. The model: the system's live reconstruction of the room -- geometry, semantics, dynamics -- the substrate of Chapter 2 and the dependency of every later chapter. The placed layer: the computed content that lives in the room's coordinates -- objects, annotations, panels, projected people -- subject to the laws of Chapter 4 and the memory of Chapter 6. The grammar: the body's channels of Chapter 5 -- hand, gaze, voice, walk -- through which the person addresses both layers in one language. The anchor: the room's remembered feature that gives placed content an address, the unit of the medium's memory. The blend: the whole construction -- one room, one model, one placed layer, one body, held together at the seam. Each term earns its keep by being load-bearing: drop one from the vocabulary and the corresponding engineering becomes invisible, which is how fields lose their disciplines -- not to refutation but to forgetting. The rest of the book keeps this dictionary as deliberately as it keeps the contract, because a field that cannot say precisely what it means cannot audit precisely what it builds.

The contract earns its final form in the buyer's hands, where it becomes three audits the product can be run through in a working afternoon, without instruments, without vendors present, and without the staged room. The first audit is the impostor audit: the three-clause scoring of the first sections, asked aloud -- does the virtual share the room's coordinates, do the two layers act on each other, is the room's model live -- and answered by observation rather than by specification, because the specification will say yes and the floor will say what is true. The second audit is the seam audit: the deliberate hunt for the boundary the marketing will not name -- what happens when the person walks out of the tracked volume, when the lamp is switched, when the furniture is moved -- the moments where the confident product either degrades honestly or reveals which clause it was never carrying; the seam's behavior under surprise is the product's real documentation, and it cannot be rehearsed by the seller who arranged the room. The third audit is the body audit: a first-time wearer, unaffiliated, unbriefed, asked to do one real task in an unprepared room while the buyer watches the body and not the screen -- because the body's account is the medium's whole believability ledger expressed in one person's posture, and the body does not read the datasheet. The three audits compress the chapter to its working size: contract, seam, body -- and the buyer who runs them in sequence holds a better definition of the product than its own marketing does, which is precisely what a definition is for.

The definition of mixed reality is no marketing posture; it is a contract three clauses deep, and those clauses are the book's skeleton. Co-location, mutual action, the live model: virtual and physical sharing a coordinate space, acting on each other, held up by a room model that changes as the room itself changes.
The continuum is a design space, not a maturity ladder; the impostors are catalogued so the buyer can audit them in an afternoon; the exclusions -- not a display, not a passthrough, not a world, not a category -- protect the boundary that makes the medium buildable. The room is a first-class citizen, believability is a budget the body keeps, and the seam is where the structure holds or fails. A builder who accepts this contract knows what to fund; a buyer who accepts it knows what to ask; and the next chapter begins where every honest anatomy begins -- with the room itself, and the model of it that every believable blend must carry.